

# Continuous Random Variables

Sourav Karmakar

# Outline

- 1 Continuous Random Variables
- 2 Expectations and Variance
- 3 Common Continuous Distributions
- 4 Quiz

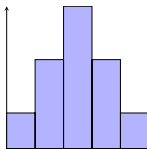
# Chapter 1: Continuous Random Variables

# Transition: Discrete to Continuous (Histograms to Curves)

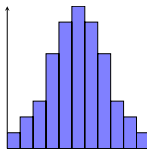
Imagine measuring the height of 10,000 people.

- As we increase the precision of our measurement (bins get narrower), the histogram approaches a smooth curve.
- In the limit (infinite precision), this curve becomes the **Probability Density Function (PDF)**.

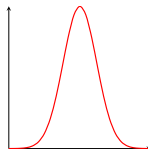
Coarse (Nearest 5cm)



Fine (Nearest 1mm)



Continuous (Infinite Precision)

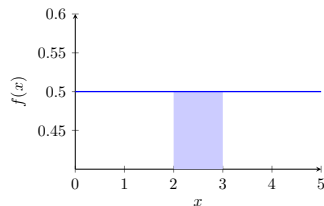


# Probability Density Function (PDF)

For Continuous Random Variables, we use a **Probability Density Function (PDF)**, denoted  $f(x)$ .

## Key Differences from PMF

- 1 **PMF (Discrete):**  $P(X = x)$  is the height of the bar.
- 2 **PDF (Continuous):** The height  $f(x)$  is **NOT** the probability.
- 3 **Area is Probability:** Probability is the **Area under the curve**.



# Calculating Probability (Integration)

To find the probability that  $X$  falls between  $a$  and  $b$ , we integrate the PDF:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

## Properties of PDF

- ①  $f(x) \geq 0$  for all  $x$ . (Density cannot be negative).
- ② Total Area = 1:  $\int_{-\infty}^{\infty} f(x) dx = 1$ .

# The Zero Probability Paradox

**Concept:** For a continuous variable, the probability of taking an **exact** value is Zero.

$$P(X = c) = \int_c^c f(x) dx = 0$$

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**Intuition: The Infinite Digital Scale** Imagine a digital scale with infinite decimal places.

- You weigh an apple.
- Is it **exactly** 100g?
- It might be 100.000000001g or 99.999999998g.
- The probability of hitting exactly 100.000... (infinite zeros) is impossible (1 outcome out of infinite possibilities).

*Conclusion: We only calculate probabilities for Intervals (e.g., between 99.0g and 101.0g).*



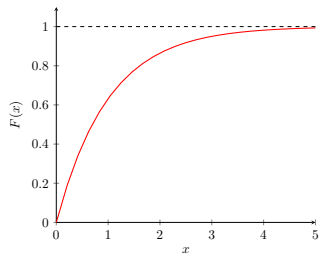
# Cumulative Distribution Function (CDF)

The CDF,  $F(x)$ , gives the probability that the variable takes a value less than or equal to  $x$ .

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

## Relationship:

- PDF is the derivative of CDF:  $f(x) = F'(x)$ .
- Prob of interval:  
 $P(a \leq X \leq b) = F(b) - F(a)$ .



# Numerical Example 1: The Linear PDF

**Problem:** Let  $X$  be a random variable with PDF:

$$f(x) = \begin{cases} 2x & \text{for } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Calculate  $P(X > 0.5)$ .

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**Solution:**

$$P(X > 0.5) = \int_{0.5}^1 2x \, dx$$

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**Solution:**

$$\begin{aligned} P(X > 0.5) &= \int_{0.5}^1 2x \, dx \\ &= [x^2]_{0.5}^1 \\ &= (1^2) - (0.5^2) = 1 - 0.25 = 0.75 \end{aligned}$$

## Numerical Example 2: The Constant PDF

**Problem:** A bus arrives randomly between 8:00 and 8:10. Let  $X$  be the minutes past 8:00. PDF:  $f(x) = 0.1$  for  $0 \leq x \leq 10$ . Find the probability the bus arrives in the first 3 minutes.

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**Solution:**

$$P(0 \leq X \leq 3) = \int_0^3 0.1 \, dx$$

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**Solution:**

$$P(0 \leq X \leq 3) = \int_0^3 0.1 \, dx$$

$$= [0.1x]_0^3$$

$$= 0.1(3) - 0.1(0) = 0.3$$

*(30% chance)*

# Chapter 2: Expectations and Variance

# Expectation (Mean) in Continuous Domain

For discrete, we summed  $x \cdot P(x)$ . For continuous, we integrate:

## Formula

$$E[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

It represents the “Center of Gravity” of the probability density.

# Numerical Example: Expectation

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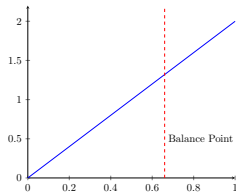
$$\begin{aligned} E[X] &= \int_0^1 x \cdot (2x) dx = \int_0^1 2x^2 dx \\ &= \left[ \frac{2x^3}{3} \right]_0^1 \end{aligned}$$

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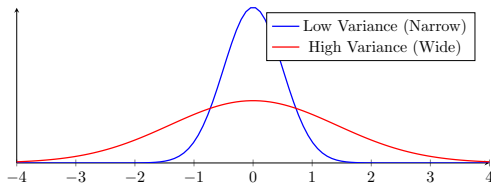
# Variance and Standard Deviation of Continuous RV

**Variance** ( $Var(X)$ ): The average squared distance from the mean. Let  $\mu = E[X]$

$$Var(X) = E[(X - \mu)^2] = \int (x - \mu)^2 f(x) dx \quad \text{OR} \quad E[X^2] - (E[X])^2$$

**Standard Deviation** ( $\sigma$ ): A measure of spread in original units.

$$\sigma = \sqrt{Var(X)}$$





# Numerical Example: Variance

**Problem:** Find variance for  $f(x) = 2x$  on  $[0, 1]$  (We know  $\mu = 2/3$ ).

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$$E[X^2] = \int_0^1 x^2 \cdot (2x) dx = \int_0^1 2x^3 dx$$

$$= \left[ \frac{2x^4}{4} \right]_0^1 = \frac{1}{2} = 0.5$$

# Numerical Example: Variance

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$$= \left[ \frac{2x^4}{4} \right]_0^1 = \frac{1}{2} = 0.5$$

**Step 2: Apply Formula**

$$\text{Var}(X) = 0.5 - (2/3)^2 = 0.5 - 0.444 = 0.055$$

# Linearity Properties

**Expectation:**  $E[aX + b] = aE[X] + b$ . **Variance:**  $Var(aX + b) = a^2Var(X)$ .

## Real World Example: Project Cost Estimation

- Let  $X$  be the time (in hours) to complete a coding task.
- Suppose  $E[X] = 10$  hours and standard deviation  $\sigma_X = 2$  hours ( $Var(X) = 4$ ).
- You hire a consultant who charges **\$50 per hour** plus a **\$100 fixed fee**.
- Cost function:  $Y = 50X + 100$ .

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- You hire a consultant who charges **\$50 per hour** plus a **\$100 fixed fee**.
- Cost function:  $Y = 50X + 100$ .

**Expected Cost:**  $50(10) + 100 = \$600$ .

**Variance of Cost:**  $50^2 \cdot Var(X) = 2500 \cdot 4 = 10,000$ .

**Std Dev of Cost:**  $\sqrt{10,000} = \$100$ .

*(Notice the fixed fee \$100 does not affect the uncertainty/spread).*

# Chapter 3: Common Distributions

# 1. Uniform Distribution

**Notation:**  $X \sim U(a, b)$

**Description:** All intervals of the same length are equally probable.

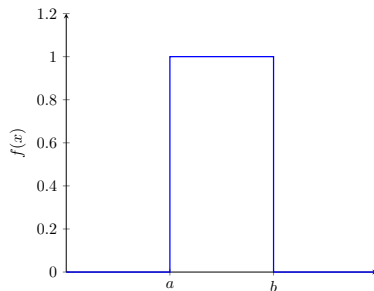
**Parameters:**

- $a$  (The Minimum): The lower bound of the distribution.
- $b$  (The Maximum): The upper bound of the distribution.

- **PDF:**  $f(x) = \frac{1}{b-a}$  for  $a \leq x \leq b$ .

- **Mean:**  $\frac{a+b}{2}$

- **Variance:**  $\frac{(b-a)^2}{12}$





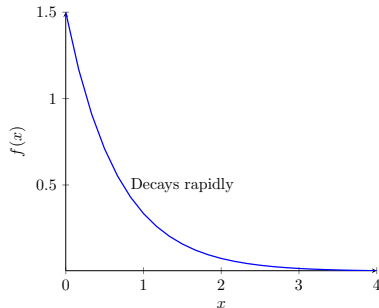
## 2. Exponential Distribution

**Notation:**  $X \sim \text{Exp}(\lambda)$

**Use Case:** Modeling time until an event (e.g., time until next customer arrives).

**Parameter:**  $\lambda$  : commonly known as the rate parameter.

- **PDF:**  $f(x) = \lambda e^{-\lambda x}$  ( $x \geq 0$ )
- **Mean:**  $1/\lambda$
- **Variance:**  $1/\lambda^2$



# Exponential: Memoryless Property

## The Property

$$P(X > s + t \mid X > s) = P(X > t)$$

**Interpretation:** The probability of waiting  $t$  more minutes is the same, regardless of how long ( $s$ ) you have already waited.

### Real Life Example:

- **Radioactive Decay:** An atom doesn't "age". The chance it decays in the next second is constant, whether it's 1 second old or 1 billion years old.
- **Customer Arrivals:** If arrivals are random (Poisson process), the time you've already waited for a customer does not change how much longer you will likely wait.

### 3. Normal (Gaussian) Distribution

**Notation:**  $X \sim \mathcal{N}(\mu, \sigma^2)$

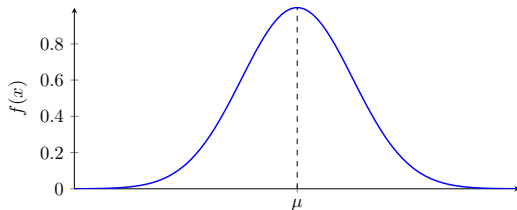
**Description:** The “Bell Curve”. Most common distribution in nature.

**Parameters:**

- $\mu$  (Mean/Location): This represents the center or peak of the “bell”
- $\sigma^2$  (Variance/Scale): This represents the “spread” of the distribution.

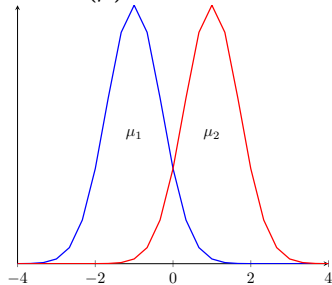
**PDF:**

$$f(x | \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

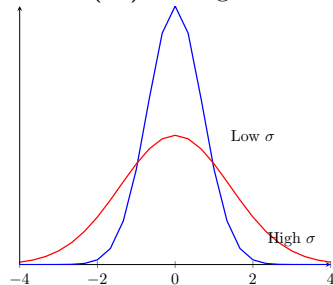


# Normal Distribution: Parameters

**Mean ( $\mu$ ):** shifts the center.



**Variance ( $\sigma^2$ ):** changes the width.

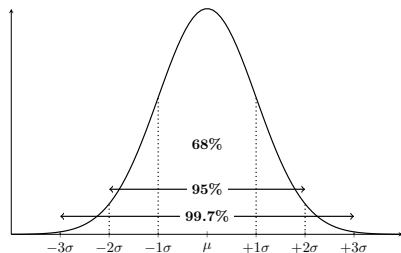


# The 68-95-99.7 Rule (Empirical Rule)

## For a Normal Distribution:

- **68%** of the data falls within **one** standard deviation of the mean ( $\mu \pm 1\sigma$ ).
- **95%** of the data falls within **two** standard deviations ( $\mu \pm 2\sigma$ ).
- **99.7%** of the data falls within **three** standard deviations ( $\mu \pm 3\sigma$ ).

*Interpretation: It is extremely rare (only 0.3% chance) for data to fall outside the  $3\sigma$  range.*



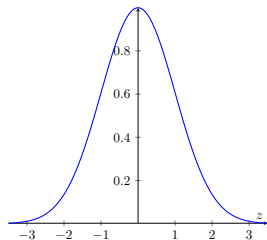
# Standard Normal Distribution ( $Z$ )

If we scale any Normal variable, we get the **Standard Normal**:

$$Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1)$$

## Properties:

- Mean  $\mu = 0$
- Std Dev  $\sigma = 1$
- PDF:  $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$
- Used for "Z-Scores" (how many sigmas away is a data point?).



## Numerical Problem 2: Manufacturing

**Scenario:** A machine cuts rods. Length  $L \sim \mathcal{N}(10\text{cm}, 0.04\text{cm}^2)$  ( $\sigma = 0.2$ ). Rods must be between 9.6 and 10.4. What % are defective?

## Numerical Problem 2: Manufacturing

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**Solution:**

- ① Limits are  $10 \pm 0.4$ .
- ② Since  $\sigma = 0.2$ , the limits are exactly  $\mu \pm 2\sigma$ .
- ③ We know 95% are inside  $\pm 2\sigma$ .
- ④ Therefore,  $100\% - 95\% = 5\%$  are defective.



## Numerical Problem 3: Inverse Problem

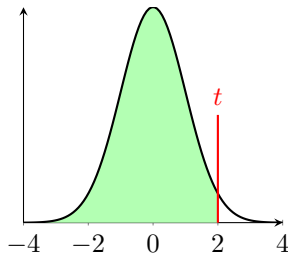
**Scenario:** Delivery time  $T \sim \mathcal{N}(30, 16)$  ( $\sigma = 4$ ). We want to promise a time  $t$  such that we are 97.5% sure we arrive before  $t$ .

# Numerical Problem 3: Inverse Problem

**Scenario:** Delivery time  $T \sim \mathcal{N}(30, 16)$  ( $\sigma = 4$ ). We want to promise a time  $t$  such that we are 97.5% sure we arrive before  $t$ .

## Solution:

- We need the cumulative area to be 0.975.
- 95% is middle. The tails are 2.5% each.
- Area to left of  $+2\sigma$  is  
50%(left half) + 47.5%(half of 95) = 97.5%.
- $t = \mu + 2\sigma = 30 + 2(4) = 38$  mins.



# Chapter 5: Quiz

## Quiz: Question 1

**Q1:** Can the value of the Probability Density Function  $f(x)$  be greater than 1?

- A) No, probability can never exceed 1.
- B) Yes, because  $f(x)$  is density, not probability.

## Quiz: Question 1

**Q1:** Can the value of the Probability Density Function  $f(x)$  be greater than 1?

- A) No, probability can never exceed 1.
- B) Yes, because  $f(x)$  is density, not probability.

**Answer: B.**  $f(x)$  represents density. For example, if  $X \sim U(0, 0.1)$ , the height is  $1/(0.1) = 10$ . The **Area** must be 1, but height can be anything.

## Quiz: Question 2

**Q2:** Calculate  $E[X]$  if  $f(x) = 3x^2$  for  $0 \leq x \leq 1$ .

A) 0.5    B) 0.75    C) 1.0    D) 0.25

## Quiz: Question 2

**Q2:** Calculate  $E[X]$  if  $f(x) = 3x^2$  for  $0 \leq x \leq 1$ .

A) 0.5    B) 0.75    C) 1.0    D) 0.25

**Answer: B) 0.75**

$$E[X] = \int_0^1 x(3x^2)dx = \int_0^1 3x^3 dx = \left[ \frac{3x^4}{4} \right]_0^1 = \frac{3}{4}$$

## Quiz: Question 3

**Q3:** For a Normal distribution, what is the probability of being more than 3 standard deviations away from the mean (in either direction)?

A) 5%    B) 1%    C) 0.3%    D) 0.0%



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A) 5%    B) 1%    C) 0.3%    D) 0.0%

**Answer: C) 0.3%** 99.7% is inside, so  $100 - 99.7 = 0.3\%$  is outside.

## Quiz: Question 4

**Q4:** The time to wait for a bus is Uniformly distributed between 0 and 20 minutes ( $U(0, 20)$ ). What is the variance?

A) 100    B) 10    C) 33.33    D) 20

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A) 100    B) 10    C) 33.33    D) 20

**Answer: C) 33.33**

$$Var(X) = \frac{(b-a)^2}{12} = \frac{(20-0)^2}{12} = \frac{400}{12} \approx 33.33$$

## Quiz: Question 5

**Q5:** You are comparing two Normal Distributions:  $X \sim \mathcal{N}(0, 1)$  and  $Y \sim \mathcal{N}(5, 100)$ . Which variable is more likely to produce a value close to its mean?

- A) X (Standard Normal)
- B) Y (Normal with Variance 100)
- C) Both are equal

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- A) X (Standard Normal)
- B) Y (Normal with Variance 100)
- C) Both are equal

**Answer: A) X.**  $Y$  has a variance of 100 ( $\sigma = 10$ ), meaning it is very spread out (flat).  $X$  has variance 1 ( $\sigma = 1$ ), meaning it is tightly clustered around the mean.

- **Continuous RVs:** Probabilities are Areas (Integrals).
- **PDF vs CDF:** Density vs Cumulative Probability.
- **Distributions:**
  - Uniform (Equal likely range).
  - Exponential (Memoryless decay).
  - Normal (Bell curve, central to statistics).
- **68-95-99.7 Rule:** Quick estimation for Normally distributed data.

Thank You!